

What are the most discussed negative effects of social media in YouTube comments which can harm Generation Z?

Abstract

Social Media is a powerful tool which can affect Generation Z in a variety of negative ways. Therefore, this report highlights the impact of social media on Generation Z through an analysis of negative comments from YouTube videos, utilizing a dataset named “Negative Comments and Metadata of YouTube videos” (Ajlal, 2024). It uses tools like TextBlob, the YouTube API, VADER, and tokenization techniques to identify common negative themes. The research finds 'sad', 'depression', and 'anxiety' as prevalent keywords, especially among Gen Z’s opinions, indicating a significant concern about social media's negative effects. This study aims to enhance awareness and encourage positive social media usage by revealing these patterns.

Introduction

This project examines the impact of social media on Gen-Z, focusing on how platforms like Instagram contribute to physical disengagement. Using the "Negative Comments and Metadata of YouTube videos" dataset (Ajlal, 2024), it aims to predict the most significant negative effects reflected in YouTube comments, particularly those related to Gen-Z opinions. Existing research in this field includes sentiment analysis which takes into account both positive and negative sentiments of social media and then compare them with each other such as extracting both positive and negative tweets from Twitter with negative tweets relating to depression (Abdou, 2020) or studying both negative and positive impacts of social media on mental health in specific countries like India (Kumar, 2023). This project specifically targets negative sentiments to understand universal social media issues, moving beyond the limitations of geographic-specific studies.

Data

The universe of the dataset is YouTube, population are the YouTube videos about negative effects of social media and set are the comments and metadata of such YouTube videos. The dataset has been created by scraping YouTube videos’ metadata and negative comments from YouTube videos of length of more than twenty words. The dataset has been categorized into comments, video title, publisher date, channel name, commenter name, comment date and video type. Using the method provided by OpenAI’s ChatGPT (personal communication, February 4, 2023) to get the YouTube API key and understand how the code works for extracting specific type of comments in a maximum way, comments are further processed into keywords for analysis. Firstly, the comments and metadata in all eight notebooks were individually scraped which were approximated to be more than 10,000. Comments were preprocessed and scraped using Text Blob python library to extract negative and relevant comments which were also conditioned to consist of more than 20 words. These comments were then combined into a single excel sheet file by concatenating them using eight google colaboratory notebooks with one of the notebooks having the code for combining all of the eight excel sheets created separately in each notebook. The time range is approximately between 2014-2023 of all videos. Positive and neutral comments have been excluded from the dataset as they are not part of the

research goal. The comments count after their collection, cleaning and preprocessing dropped down to 1621 with a mean negative sentiment of -0.23 showing that most of the extracted comments are at a great extent, negative. The standard deviation is also around 0.57 which means that there's not a lot of spread in the data and most the values (sentiment scores) are similar. Focusing on video type, there are four them: GenZ's Opinions, Ted Talks, short film and a research-based video. Most of the comments are from the type GenZ's opinions. The detailed summary statistics has been provided in one of the notebooks (Ajlal, 2024).

Methods

YouTube API

This is used to access the video content, comments and metadata of the set for this project and to seamlessly scrape those details.

TextBlob (Python Library)

This has been used to extract negative and relevant comments from YouTube videos by specifying a condition using sentiment polarity of less than 0 which helps in scraping comments with negative sentiment scores.

Vader (Sentiment Analysis Tool)

Firstly, Vader was used to analyze all of the comments in order to do a complete scan of each sentence and identify keywords which are negative in general. Since Vader already knows how to scrape all texts for particular sentiments, there's also no need to train the dataset. Besides, manual identification of keywords which are related to negative effects of social media is harder as humans cannot easily identify all of them from a large dataset (OpenAI's ChatGPT, personal communication, February 29, 2023). Additionally, this tool has also been used to calculate sentiment scores for all comments. The lambda function is used to take input each comment as 'x'. Polarity scores, which range from -1 (most negative sentiment) to 1 (most positive sentiment) – while 0 being neutral – are used to create a dictionary of all comments, showing what numerical sentiment score have they received. These scores have been further categorized into negative (if < 0) or positive (if > 0) and neutral otherwise. The average sentiments have also been calculated by video type. Similarly, Vader has also been used to combine each negative keyword in a comment with the video type it is situated in to perform further analysis about what kind of videos can have what kind of negative keywords associated with it to imply evaluations. Polarity score is also used here to create a dictionary of words which are negative (< 0), assigning them to their relevant video type.

Tokenization (Natural Language Toolkit)

Tokenization has been used here to split the comment texts into tokens and in this particular research, it is combined with Vader to extract relevant keywords from the already preprocessed comments. Further, bigrams have also been scraped to take into account issues like an appearance of a keyword having a changed meaning when used with another word such as a comment having words like 'there's no depression like feeling from social media' which would be opposite of what this research is looking

for. Hence, taking this into account ensures the results are more appropriate and do not conflict with the results obtained from unigrams.

Histograms:

Results such as unigram keywords, keywords specifically used in the video type of GenZ's opinions and bigrams are constructed through before and after filtered by keywords histograms as they can simply show the frequency of all of the common and relevant unigrams/bigrams. They can help in analysis of which keywords have the highest frequencies and which ones are considered the least important.

Distribution plot:

This plot has been used to show sentiment scores of negative extracted comments by using Vador. The sentiment analyzer assigns a numerical value to each comment using lambda.

Results:

The major findings can be interpreted from filtered negative unigrams and filtered negative unigrams by a video type (GenZ's opinions) which both coincidentally show that the keyword 'sad' is the one with highest frequency from both histograms for both of these analyses (Figure 1 and 2). This analysis was possible after filtering irrelevant common keywords which were not directly related to negative effects of social media. Figures 4 and 5 show how keywords like 'bad', 'no' and 'hate' have the highest frequencies before they were filtered out to better interpret the results. Therefore, these results answer the research question of the most discussed negative effects in YouTube comments', showcasing that these are the negative effects people possibly have as most of them type their own personal experiences.

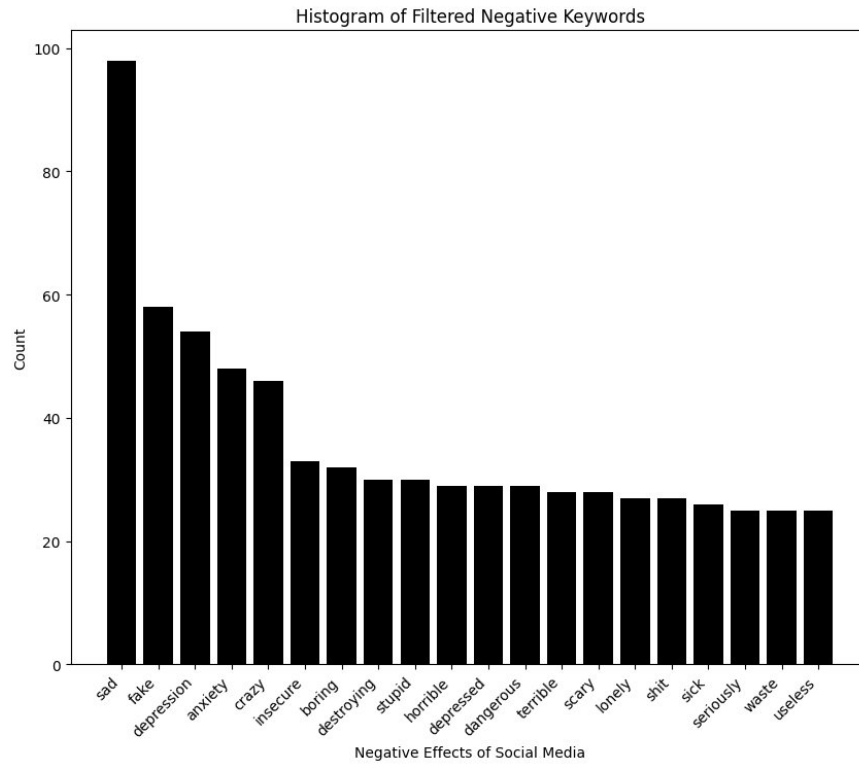


Figure 1: The histogram shows the count of keywords of 20 most repeated keywords as negative effects of social media with the count of the keyword 'sad' reaching nearly 100 followed by keywords like 'fake' and 'depression' having a count of more than 50.

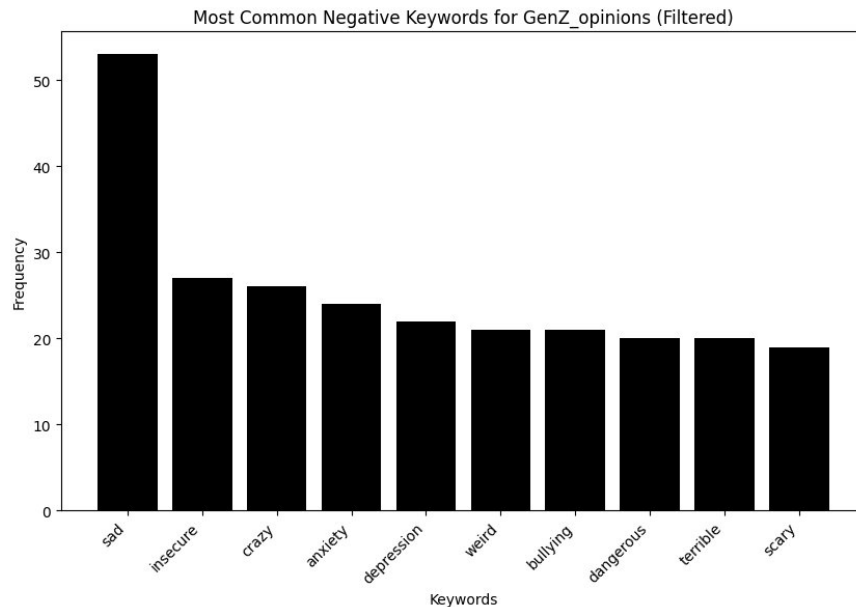


Figure 2 : The histogram shows negative and filtered keywords for a video type of category of GenZ's opinions with the keyword 'sad' still having the highest frequency count of more than 50 words. It is followed by 'insecure' and 'crazy' with a count of around 25.

However, before this research, to make sure that most of the extracted comments have a negative sentiment, a distribution plot of negative sentiment scores was created (Figure 3) which proves that most of the comments extracted are indeed negative. This shows that the TextBlob python library has scraped most of the negative and relevant comments from all of the YouTube videos.

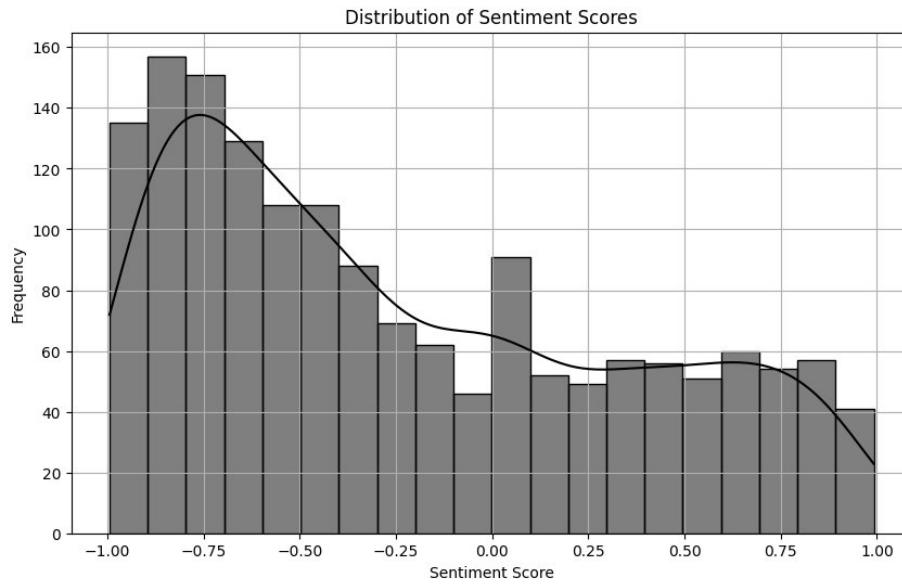


Figure 3: The distribution plot shows most of the comments filtered and extracted have negative sentiment scores with the sentiment score range of -1 to -0.75 having the highest frequency of comments (more than 120).

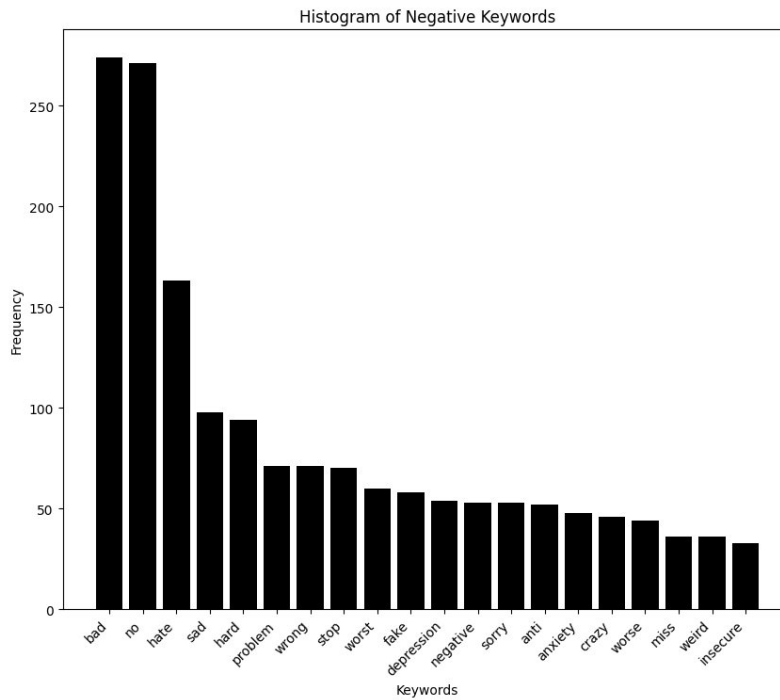


Figure 4: The histogram shows unfiltered negative keywords extracted using the sentiment analyzer with the keyword 'bad' and 'no' have the highest frequency of more than 250 words.

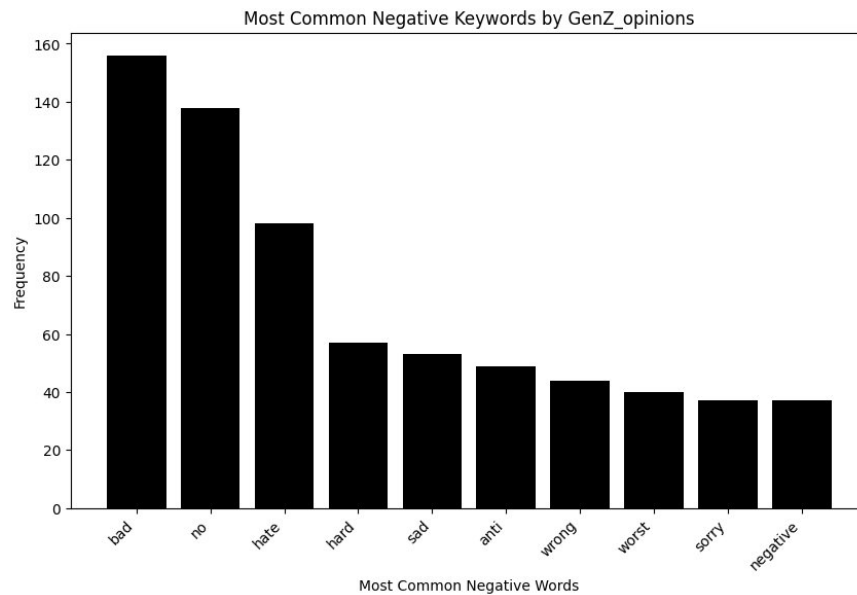


Figure 5: This histogram shows similar results but for a video type category of GenZ's opinions with the keywords 'bad' and 'no' having the highest frequencies of around 160 and 140 respectively.

To make sure that unigrams are not providing invalid results as they would not take into account bigrams like 'no depression' or 'not saddening', histograms of both unfiltered and filtered bigrams were created (Figure 6 and 7) and confirmed that even after filtering some irrelevant bigrams, the remaining most common bigrams do not show opposite results of what unigrams show but still reflect negative sentiments of users. From the comparison of both figures, it can be seen that figure 7 has comparatively more relevant words than figure 6. The highest frequency of the bigram 'anti social' shows how most users could possibly have been supporting movements or practices to stop using social media.

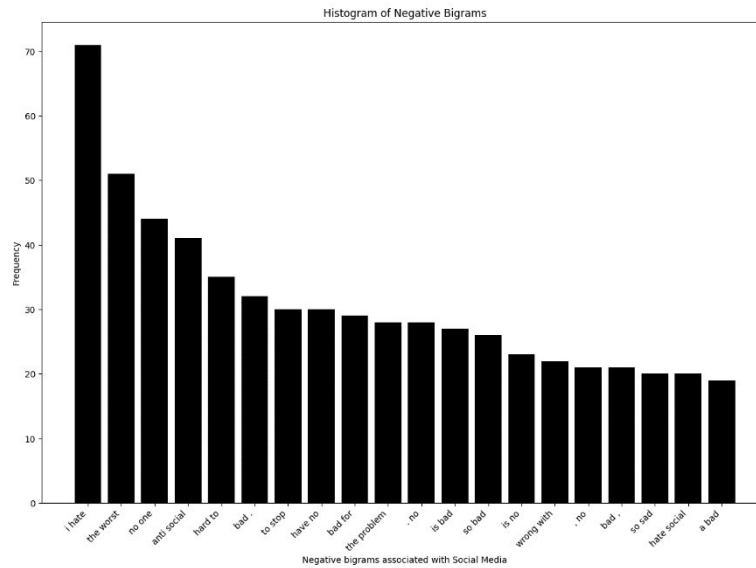


Figure 6: The histogram shows unfiltered negative bigrams extracted from comments with the highest frequency of bigrams is of 'i hate' with a count of more than 70. This is followed by the bigrams 'the worst' and 'no one' with frequencies of more than 40.

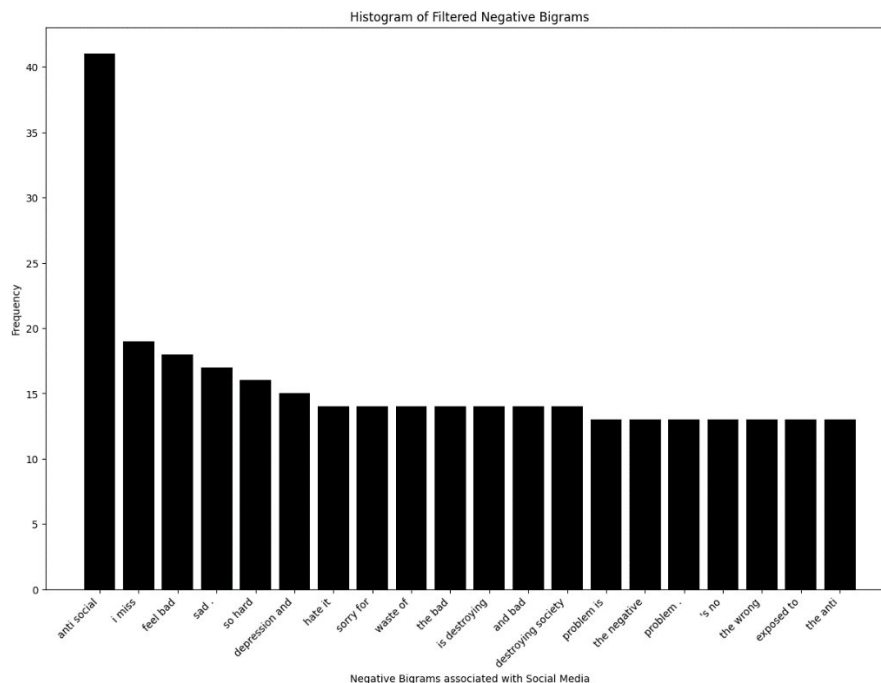


Figure 7: The histogram shows filtered negative bigrams with the bigram 'anti social' having the highest frequency (more than 40). This is followed by bigrams like 'i miss' and 'feel bad' with frequencies of more than 15.

Besides, Table 1 of average sentiments of each video type has been created which shows that even after comments are analyzed by video type, most of the comments in them have negative sentiments which show that the selection of videos for specifically analyzing negative sentiments is appropriate.

Video Type ▾	Average Sentiment ▾
GenZ_opinions	-0.19
Research	-0.36
Short_Film	-0.22
TED_Talks	-0.22

Table 1: The table shows average sentiments sorted by video type. All of them have negative average sentiments. Video type Research has the highest average negative sentiment (-0.36).

Lastly, for further sensitivity analysis, histograms (Figures 2, 5, 8, 9, 10, 11, 12 and 13) of negative sentiments were created by video type to show what are the most common negative sentiments for each video type. These were also filtered to remove the irrelevant common keywords. For Ted Talks, Short Film and research, the keywords ‘anxiety’, ‘fake’ and ‘destroying’ are the most common negative sentiments (Figure 9, 11 and 13).

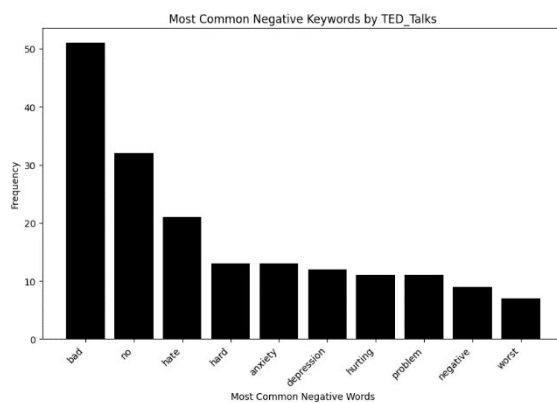


Figure 8

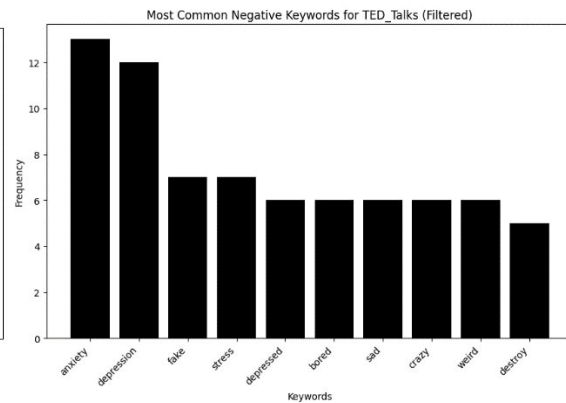


Figure 9

Figures 8 and 9: These histograms show results before and after filtering for negative keywords by the video type TED Talks.

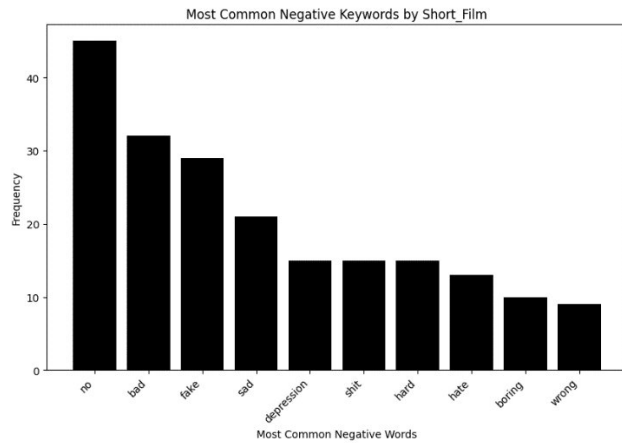


Figure 10

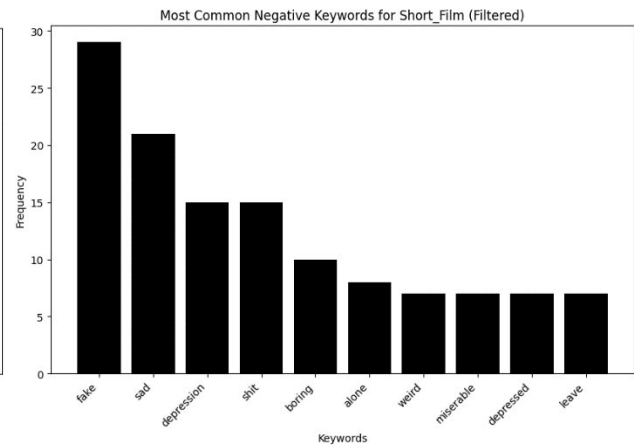


Figure 11

Figures 10 and 11: These histograms show results before and after filtering for negative keywords by the video type Short Film.

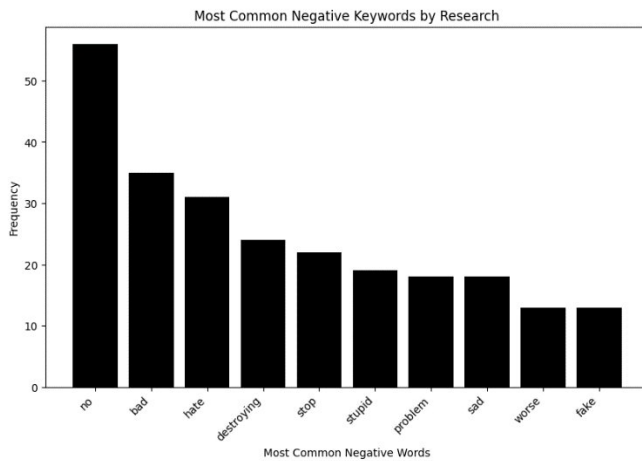


Figure 12

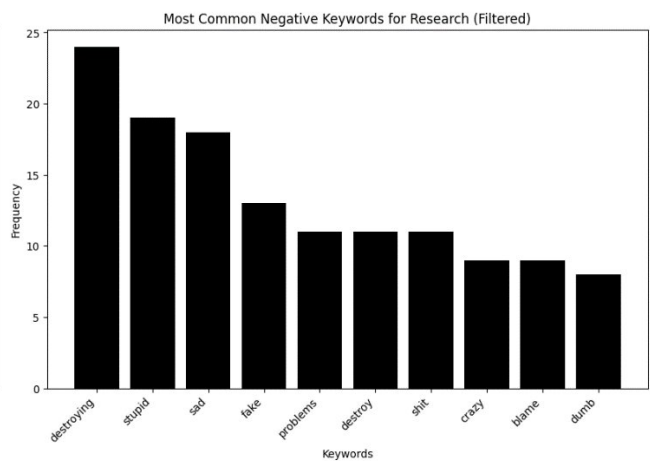


Figure 13

Figures 12 and 13: These histograms show results before and after filtering for negative keywords by the video type Research.

Conclusion

This project analyzes comments from eight YouTube videos (2014-2023) to identify common negative social media effects, particularly on Generation Z. By examining keywords in negative comments, it reveals prevalent issues among Gen-Z users. Keywords like ‘sad’, ‘fake’, and ‘depression’ were frequently found, suggesting a focus for potential social interventions. Although it provides insights, the research's scope is limited by the dataset's size and the specific user base of the analyzed channels. Future research could expand the analysis to other platforms and a broader comment range to understand and mitigate negative social media impacts more effectively.

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